Predictive Model Plan

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1. Model Logic (Generated with GenAI)

The predictive framework is built to forecast credit card account delinquency by outputting a precise risk probability for each customer entry. The system handles raw data imbalances and applies operational guardrails identified during the exploratory analysis before generating predictive outputs.

Workflow Framework:

- **Data Ingestion & Preprocessing:** Automatically handles missing entries via median and mean imputation rules, while constraining credit utilization fields to a hard mathematical ceiling of 1.0.
- **Behavioral Aggregation:** Quantifies historical customer performance by encoding the 6-month categorical status vectors into a continuous risk tracking variable.
- **Ensemble Processing:** Passes the compiled feature matrix into gradient boosted decision trees to determine the final probability of delinquency.

```
# Implementation Code for Credit Delinquency Prediction Pipeline
import pandas as pd
from sklearn.pipeline import Pipeline
from sklearn.compose import ColumnTransformer
from sklearn.impute import SimpleImputer
from sklearn.preprocessing import StandardScaler, OneHotEncoder
from xgboost import XGBClassifier

def initialize_credit_model():
    # Construct numeric transformer using median imputation strategy
    num_pipeline = Pipeline(steps=[
        ('imputer', SimpleImputer(strategy='median')),
        ('scaler', StandardScaler())
    ])

    # Construct categorical transformer for text fields
    cat_pipeline = Pipeline(steps=[
        ('encoder', OneHotEncoder(handle_unknown='ignore',
sparse_output=False))
    ])

    # Combine feature transformers
```

```
    preprocessor = ColumnTransformer(transformers=[
        ('num', num_pipeline, ['Age', 'Income', 'Credit_Score',
'Credit_Utilization', 'Loan_Balance', 'Debt_to_Income_Ratio',
'Account_Tenure', 'Unfavorable_Payments']),
        ('cat', cat_pipeline, ['Employment_Status', 'Credit_Card_Type',
'Location'])
    ])

    # Establish complete workflow architecture with class balancing
    full_pipeline = Pipeline(steps=[
        ('preprocessor', preprocessor),
        ('classifier', XGBClassifier(
            n_estimators=150,
            max_depth=4,
            learning_rate=0.05,
            scale_pos_weight=5.25,
            random_state=42
        ))
    ])
    return full_pipeline
```

2. Justification for Model Choice

An XGBoost Ensemble Model was selected for Geldium Finance's delinquency forecasting system over traditional linear approaches to optimize accuracy while maintaining operational transparency. Standard models like Logistic Regression struggle to automatically detect complex, multi-variable financial interactions, such as a high debt-to-income ratio becoming exponentially more hazardous when paired with high credit utilization. XGBoost identifies these interaction thresholds natively without requiring manual variable creation.

To balance the complex tree logic with corporate transparency and regulatory requirements, the model will be deployed alongside SHAP (Shapley Additive exPlanations) values. This provides clear explanations for individual risk scores, ensuring that automated credit decisions are interpretable and fully auditable by risk management teams.

3. Evaluation Strategy

The model's performance will be evaluated using a comprehensive suite of metrics tailored to handle imbalanced credit risk data:

- **F1-Score & Recall:** Rather than relying on simple accuracy—which can be misleading when delinquent accounts are a minority—the system prioritizes Recall to minimize missed flags on high-risk accounts, balanced by the F1-Score.

- **ROC-AUC Score:** Measures the model's global capacity to differentiate high-risk accounts from low-risk accounts across varying decision cutoffs. We target an operational minimum score of ≥ 0.82.
- **Stratified 5-Fold Cross-Validation:** Ensures structural reliability and guards against overfitting by verifying that model performance remains stable across multiple balanced data splits.

Ethical considerations and fairness are integrated directly into the deployment plan. The system will be audited for bias across sensitive attributes, such as geographical location and age groups, using the **Disparate Impact Ratio** and **Equalized Odds** frameworks. A minimum metric target of 0.80 will be enforced to ensure fair treatment. If any structural bias is discovered, classification thresholds will be adjusted dynamically during post-processing to maintain equitable risk scoring across all customer groups.